MOHAMMED SABITH

Structural Engineer | RC High-Rise | Industrial Steel | Structural Retrofitting

Doha, Qatar | +974 5037 1817 | sabithismail6679@gmail.com | linkedin.com/in/sabithismail | sabith-eng.vercel.app

Valid QID (Transferable) | Qatar Driving Licence

PROFESSIONAL SUMMARY

Structural Design Engineer in Doha working on reinforced concrete and structural steel buildings, from villas to a 2B+G+12 commercial building and industrial warehouses. Designs in ETABS, SAFE and STAAD.Pro to ACI 318, ASCE 7, AISC and QCS, and writes Python tools to automate the repetitive parts. UPDA engineer registration in progress.

WORK EXPERIENCE

Structural Design Engineer | Sketch Engineering Consulting | *Doha, Qatar* *Feb 2025 – Present*

- Produced full structural design packages for 50+ villas and 10 commercial buildings, including several villas with post-tensioned (PT) slabs spanning up to 15 m.
- Modelled the shear walls and flat slabs of a 2B+G+12 commercial building in ETABS and SAFE, and optimised its raft foundation for high soil bearing pressures.
- Developed a Python plugin (CSI API) that saves 2–3 hours per project on combined footing modelling in SAFE. It reads ETABS column reactions, finds the load resultant centre and places the footing from AutoCAD shapes.
- Built a Python tool that produces AutoCAD column layout and detail drawings in seconds, replacing about 1.5 hours of manual drafting each time. It tells columns from walls by code length-to-thickness ratios and tags them to the grid. Both tools are used by 5 engineers.
- Designed a 90 m × 47 m steel factory in STAAD.Pro (22 m and 25 m clear-span portal frames, crane runway beams, 18.7 m eaves), plus warehouses and pipe supports with IDEA StatiCa connections.
- Carried out structural audits of 2 buildings and designed the retrofit for a G+11 residential block with failing slabs and columns (CFRP, concrete jacketing, steel beams).
- Supervised BIM modellers and CAD technicians, and checked architectural and MEP models against structure.
- Checked complex elements with hand calculations before approving shop drawings.
- Wrote structural assessment reports for clients and answered site queries with supporting calculations.

Structural Designer (Apprenticeship) | Techmaghi *May 2024 – Nov 2024*

- Took multi-storey reinforced concrete buildings from analysis model to reinforcement detailing, including seismic loads, ductile shear walls, flat slabs and raft foundations.

EDUCATION

Bachelor of Technology in Civil Engineering *Mar 2020 – Mar 2024*

MEA Engineering College (APJ Abdul Kalam Technological University) | *Kerala, India*

PROFESSIONAL REGISTRATION & DEVELOPMENT

- **UPDA / MMUP Engineer Registration (Qatar):** application submitted, in progress
- **IStructE and ICE:** working towards Graduate Membership, with the aim of chartership

CERTIFICATIONS & TRAINING

Structural Design and Analysis (STAAD.Pro) | Bentley Systems and Arabian Infotech Training Institute *Oct 2025*

- Steel design in STAAD.Pro, PROKON and IDEA StatiCa: ASCE 7 loads, AISC and Eurocode connections, footings.

TECHNICAL SKILLS

- **Main Software:** ETABS, CSI SAFE, STAAD.Pro, Autodesk Revit
- **Working Knowledge:** IDEA StatiCa, AutoCAD, Tekla Structures, SAP2000, Robot Structural Analysis, PROKON
- **Programming & Automation:** Python, CSI API (ETABS, SAFE), AutoCAD automation
- **Design Areas:** Reinforced Concrete Design, Structural Steel Design, Portal Frames, Post-Tensioned (PT) Slabs, Flat Slabs, Shear Walls, Raft and Pile Foundations, Seismic and Wind Analysis, Structural Retrofitting
- **Codes & Standards:** ACI 318, ASCE 7, AISC, Eurocodes, Qatar Construction Specifications (QCS)
- **Languages:** English (Professional), Malayalam (Native), Hindi (Fluent), Arabic (Basic)